

Please review the syllabus in regards to make-ups and tardiness for exams. Stuff you should know:

Sets

- Basic notation/operations, e.g., $A \cap B$, $A \cup B$, $A - B$, $A \times B$ etc.
 - Note: it makes no sense to apply these operations to elements. For example, we can't write $a \cap A$ if a is an element in A . We can write $\{a\} \cap A$.
- To show that two sets are equal, $A = B$ show $A \subseteq B$ and $B \subseteq A$.
 - To show that $A \subseteq B$, let $a \in A$ and then show that $a \in B$.
- If $|A| = n$ then A has 2^n subsets, including $\emptyset$.

Relations

- Important definitions: reflexive, symmetric, transitive, and equivalence:
- To show that $\mathcal{R}$ is one of these, apply the definition to arbitrary x, y, z .
- To show that $\mathcal{R}$ is not one of these (usually) use a counterexample.

Functions

- Important definitions: one-to-one, onto, bijection, and group homomorphism.
- Show a function is/isn't one-to-one, onto, or bijective.
- Pigeon-hole Theorem.
- Function composition.

Groups

- Important definitions: group, subgroup, and cyclic.
- Prove/disprove that a set is a subgroup.
- Apply Lagrange's Theorem.
- You will not have to prove anything involving cosets.
- Review the HW results on the groups S_n and D_n .
- Find nontrivial subgroups of a given group.

Rings

- Important definitions: rings
- Show that a given set is a ring.
 - If it is a known abelian subgroup, just state that and then show multiplicative associativity and distribution.

Some general notes

- Review HW problems. many of the exam questions are taken or modeled after HW problems.
- There will be some multiple choice problems.
 - ex. Let G be an abelian group and H be a subgroup of G . Which of the following statements are correct?
 - (a) H is cyclic.
 - (b) H is an abelian group.
 - (c) H is a finite group.
 - (d) H is a non-abelian group.